

Python Basics

INTRODUCTION TO PYTHON

- Python is a high-level, interpreted programming language
- Python has a large standard library
- Python is cross-platform
- Python uses indentation to define code blocks
- Python has a minimal number of keywords

VARIABLES AND DATA TYPES

- Variables are names given to values
- Python has several built-in data types, including integers, floats, strings, boolean, lists, and tuples
- You can assign a value to a variable using the assignment operator (=)
- You can perform basic operations on variables, such as addition, subtraction, multiplication, and division
- Variables can be used to store and manipulate data in a program

CONTROL STRUCTURES

- Conditional statements are used to control the flow of a program
- The if-else statement is used to execute a block of code if a condition is true, and another block of code if the condition is false
- For loops are used to execute a block of code repeatedly for a specified number of times
- While loops are used to execute a block of code repeatedly while a certain condition is true
- The break statement is used to exit a loop prematurely, while the continue statement is used to skip to the next iteration of a loop

FUNCTIONS

- Functions are blocks of code that can be called multiple times from different parts of a program
 - Functions are used to organize code, reduce repetition, and make programs more modular
 - A function is defined using the `def` keyword followed by the function name and a list of parameters in parentheses
 - Function arguments can be positional or keyword arguments
 - A function can return a value using the `return` statement
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MODULES AND PACKAGES

- Modules are pre-written code that can be imported into a program to provide additional functionality
 - Modules are used to organize code, reduce repetition, and make programs more modular
 - A module can be imported using the `import` statement
 - A module can be created by saving a Python file with a `.py` extension
 - Packages are collections of related modules
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